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(54) **VIRTUAL BACKGROUND SELECTION
BASED ON COMMON MEETING DETAILS**

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H04L 12/18 (2006.01)

(52) **U.S. Cl.**
CPC **H04N 7/157** (2013.01); **H04L 12/1818**
(2013.01)

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CPC H04N 7/157; H04N 7/147; H04N 7/15;
H04L 12/1818; H04L 12/1827

See application file for complete search history.

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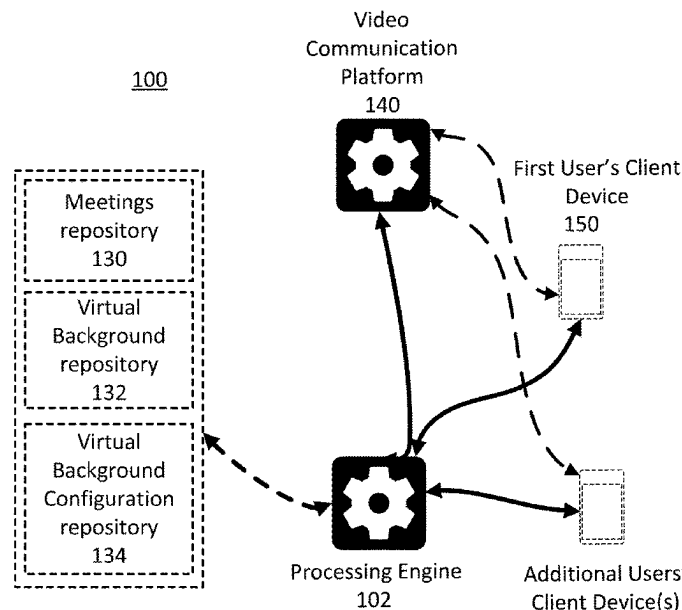
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(57) **ABSTRACT**

Usage of the same virtual background during video communications associated with multiple meetings is identified. Common information details in respective information details of the multiple meetings are identified. A virtual background selection rule that associates the common information details of each of the multiple meetings with the same virtual background is automatically generated. A determination is made that information details of a planned meeting matches the common information details. In response to determining that the information details of the planned meeting matches the common information details, the same virtual background of the virtual background selection rule is used for display during the planned meeting.

20 Claims, 14 Drawing Sheets



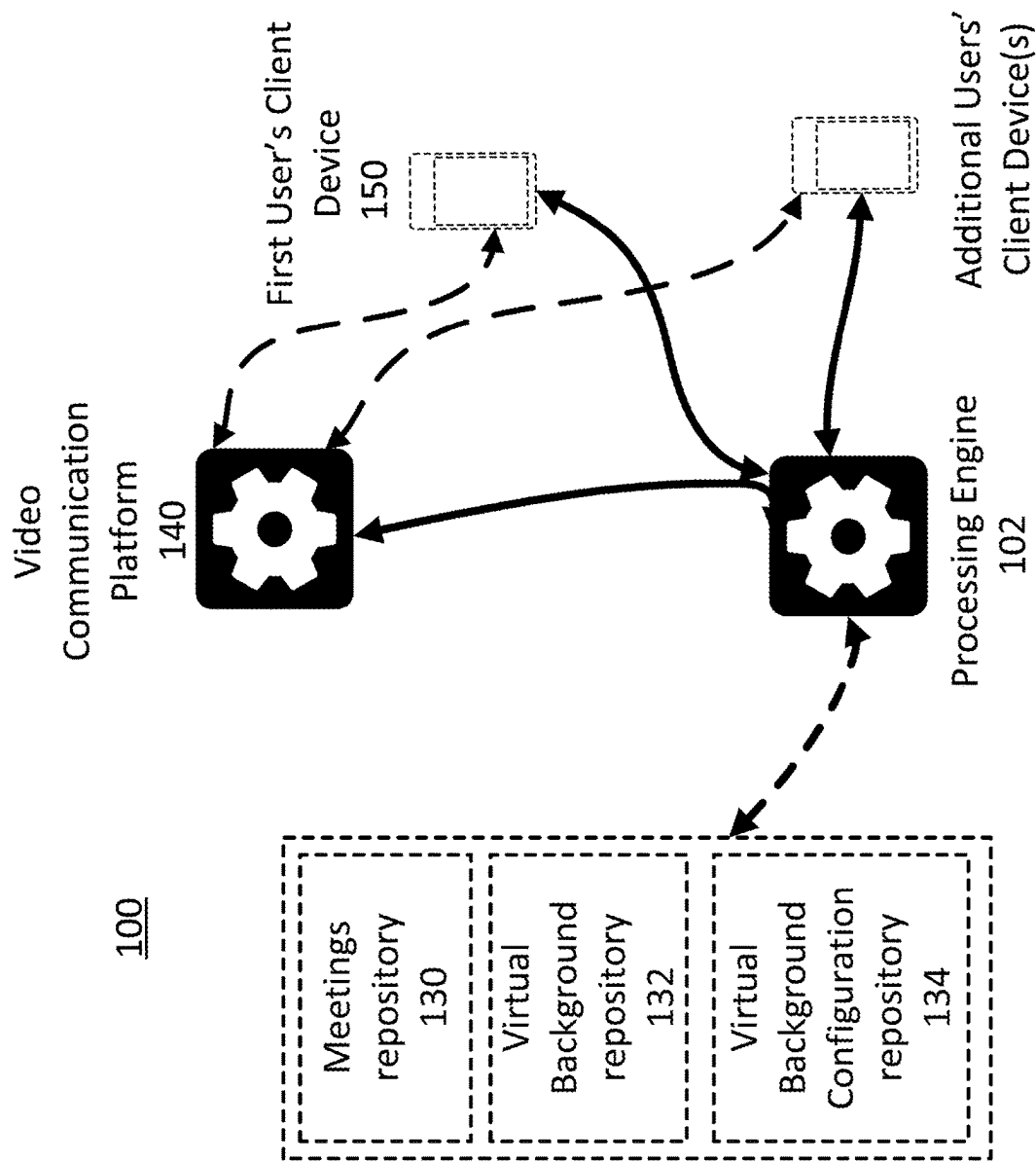


Figure 1A

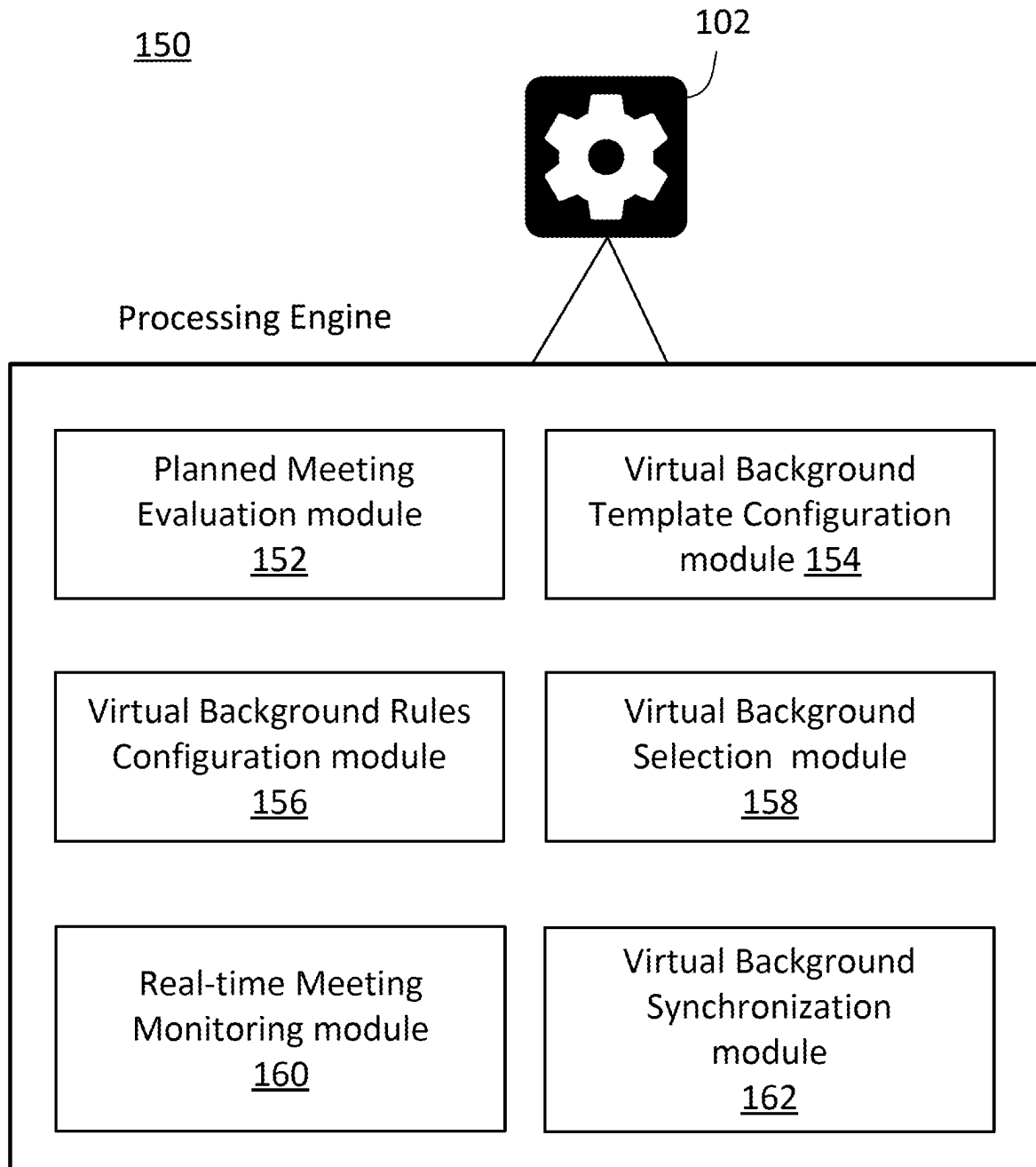
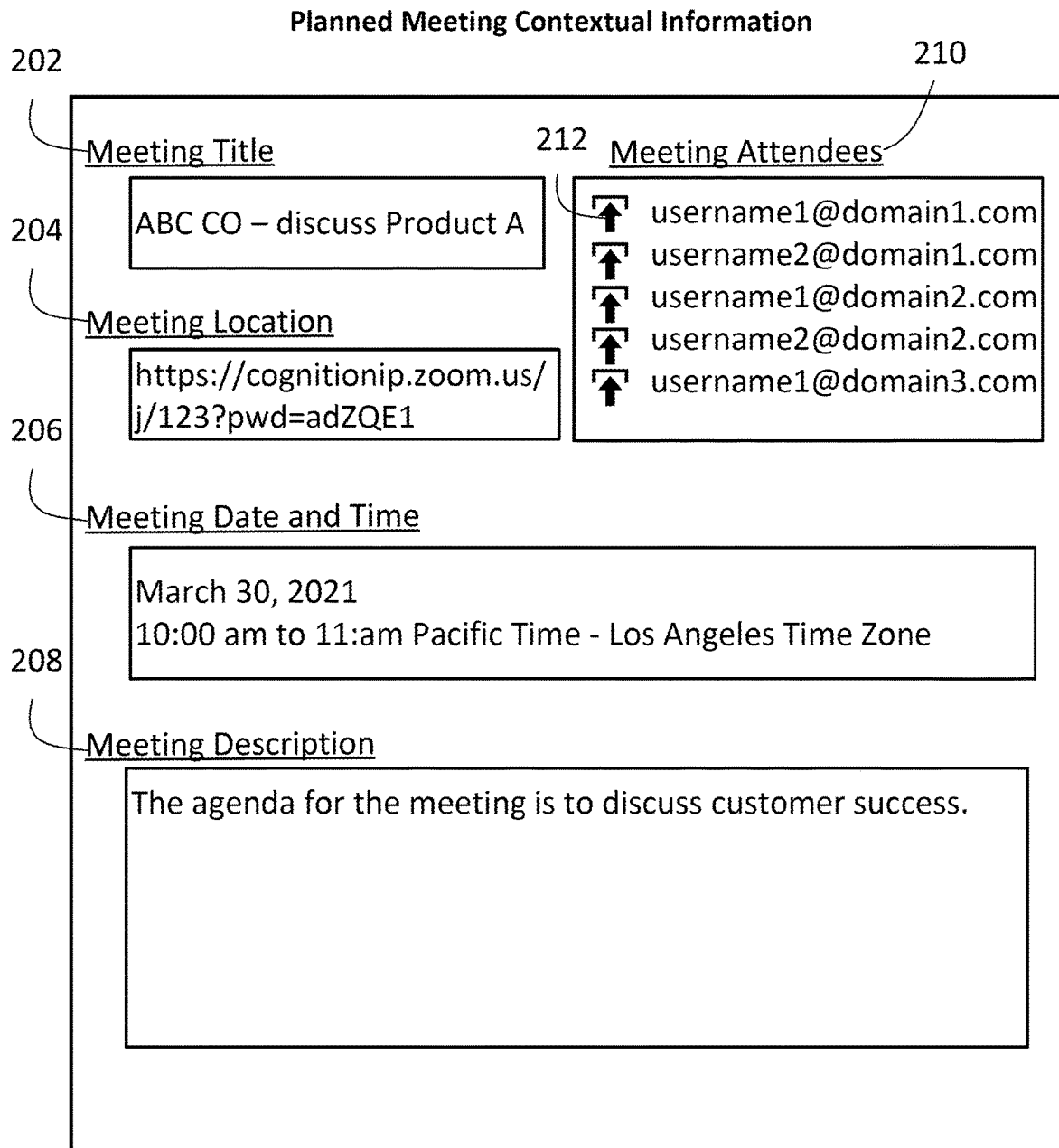


Figure 1B

**Figure 2**

**Virtual Background Selection
Using Preconfigured Rules**

300

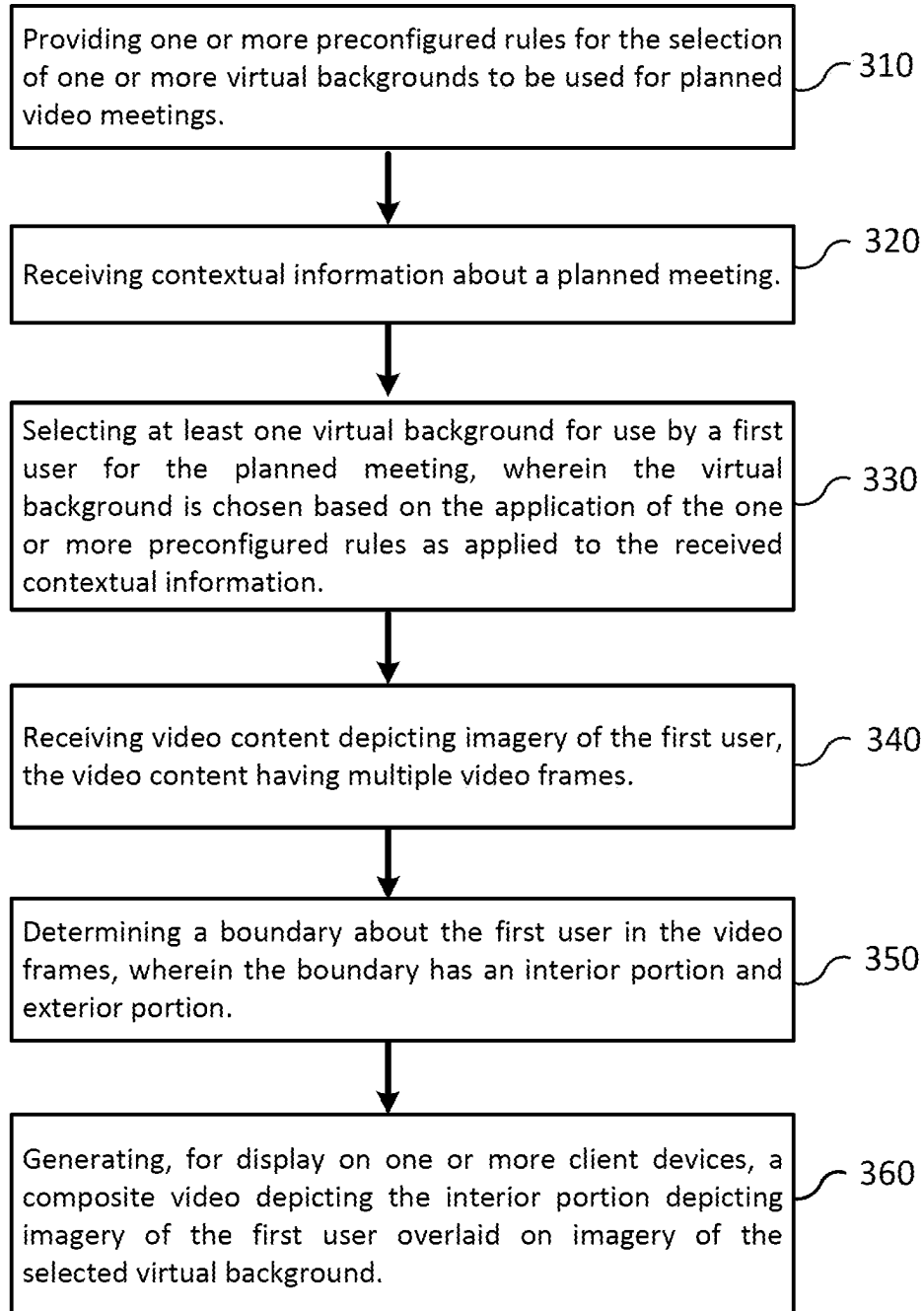
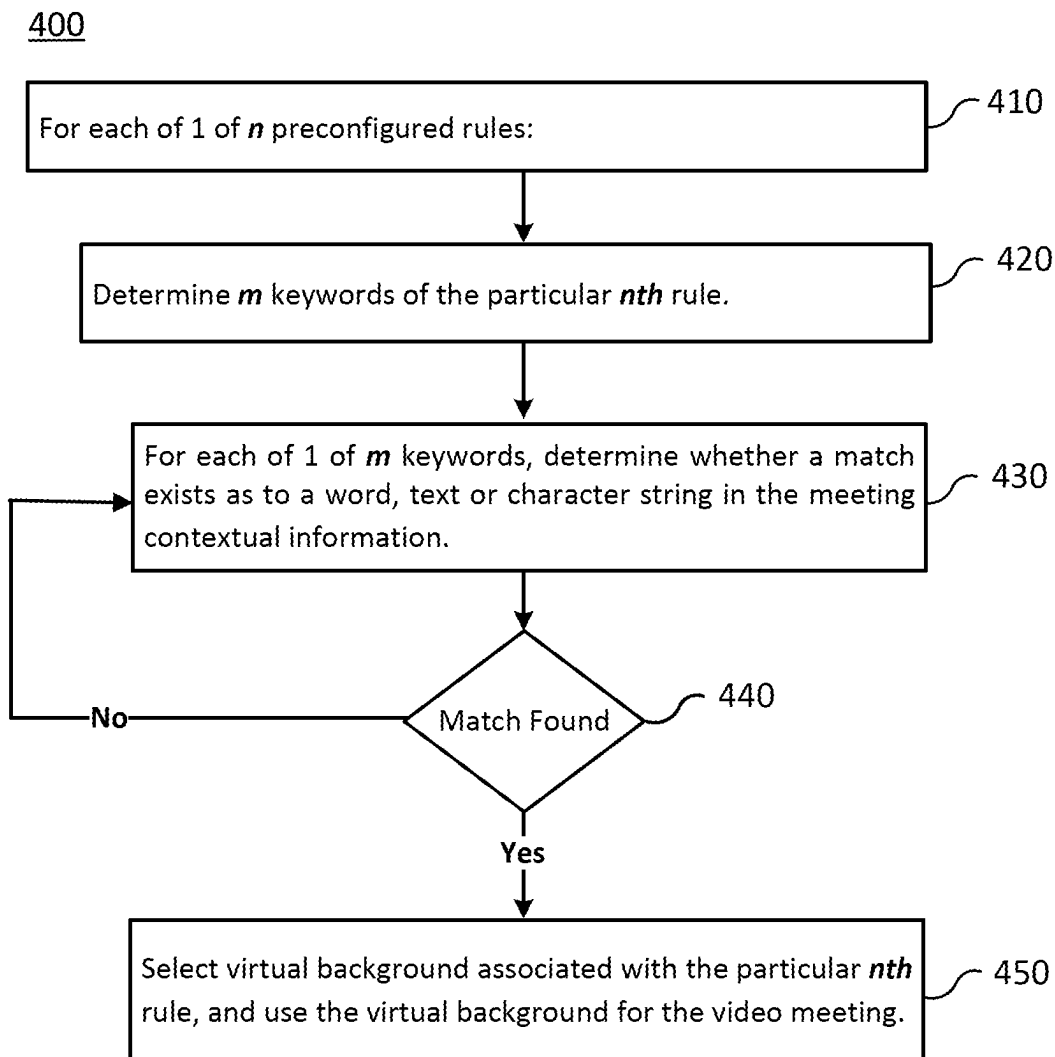


Figure 3

Preconfigured Rules Evaluation Process

**Figure 4**

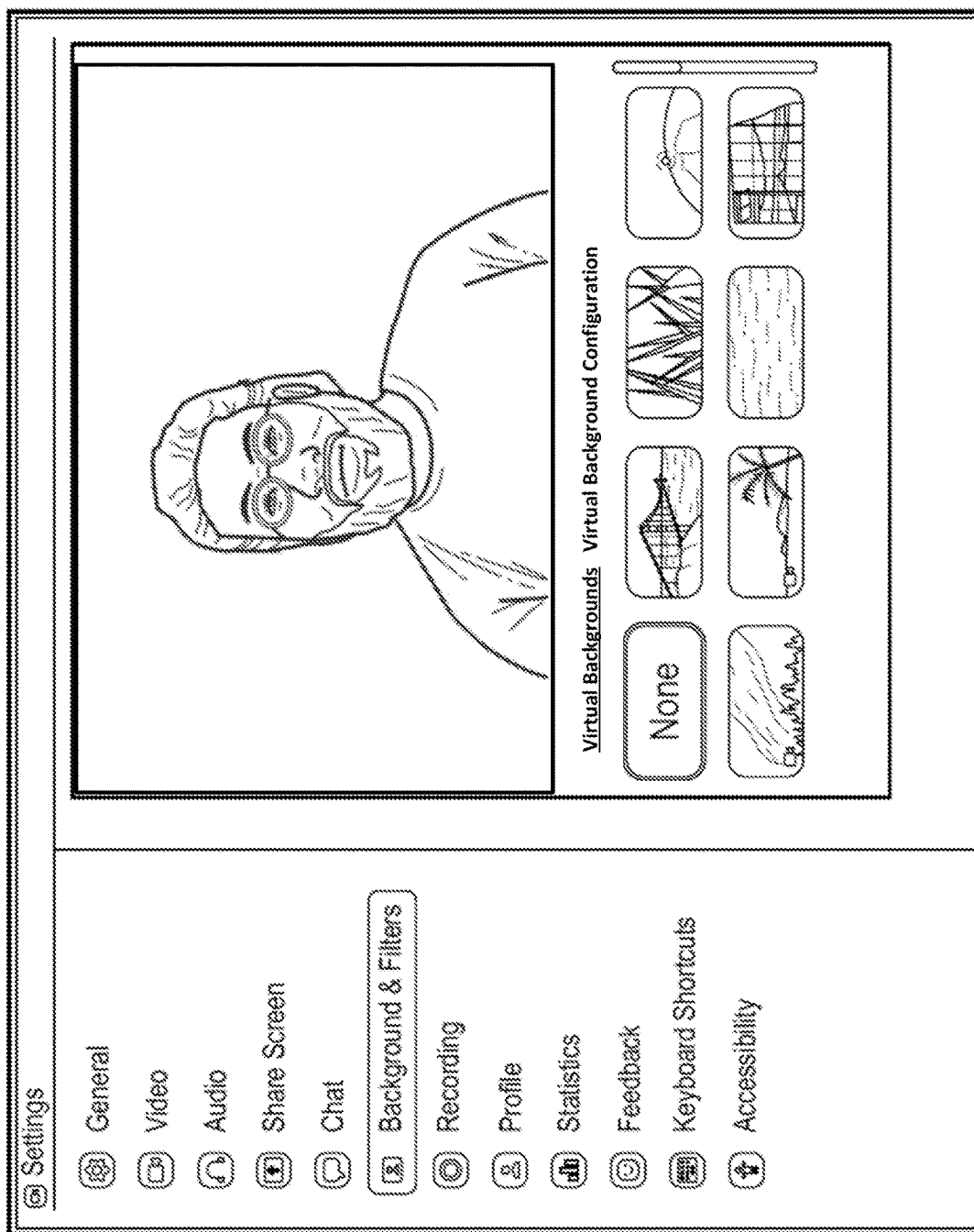


Figure 5A

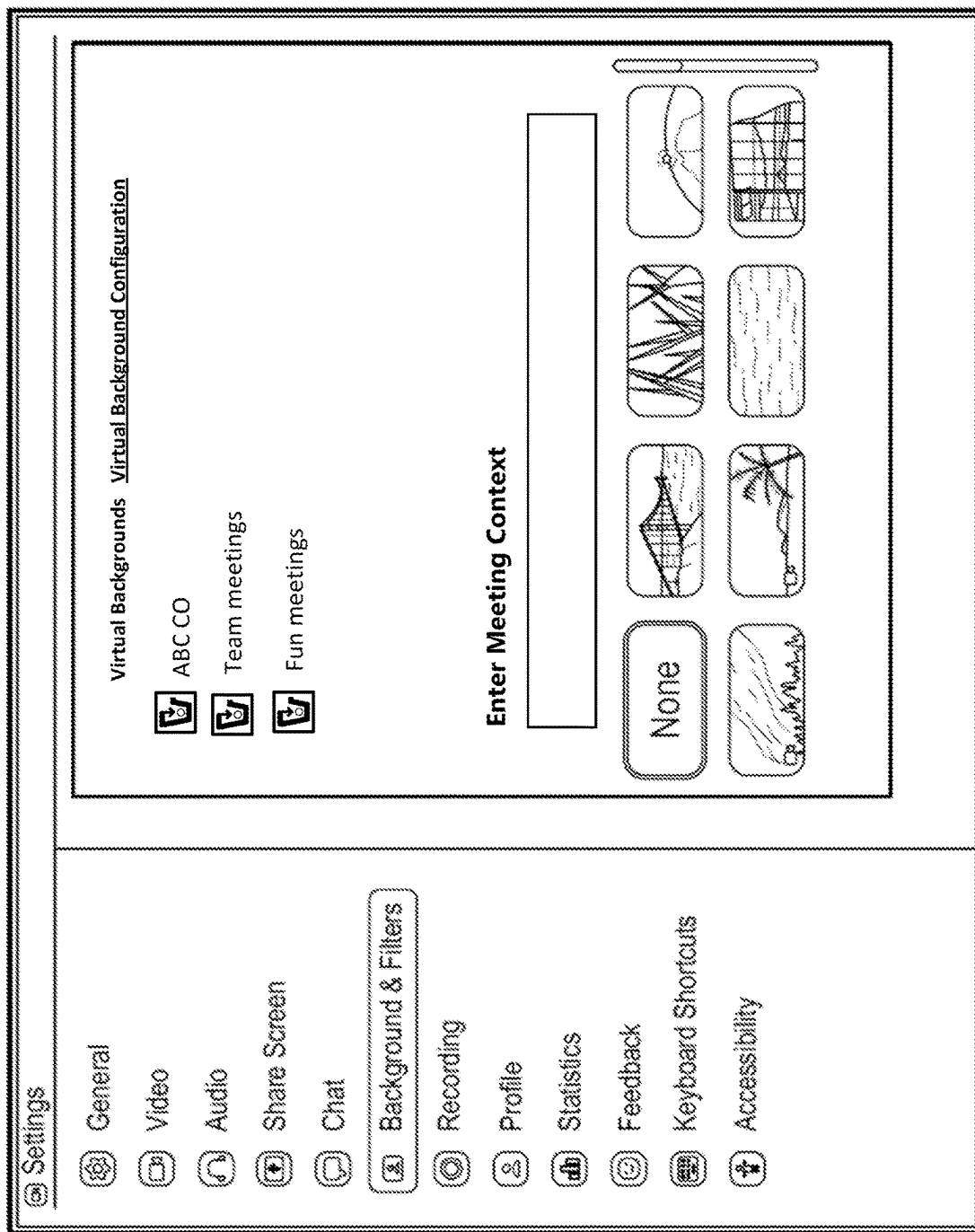


Figure 5B

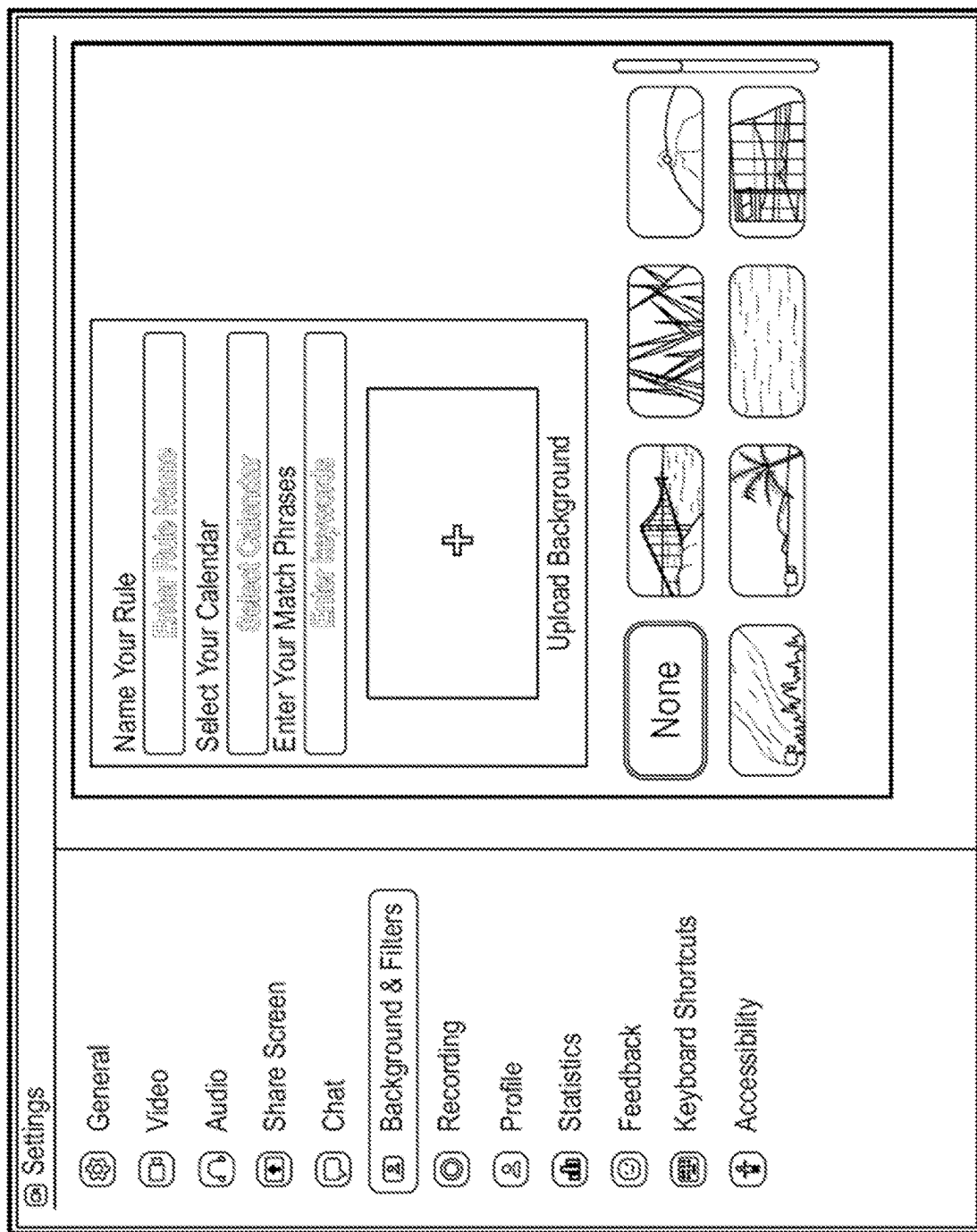


Figure 5C

Virtual Background Template Configuration

600

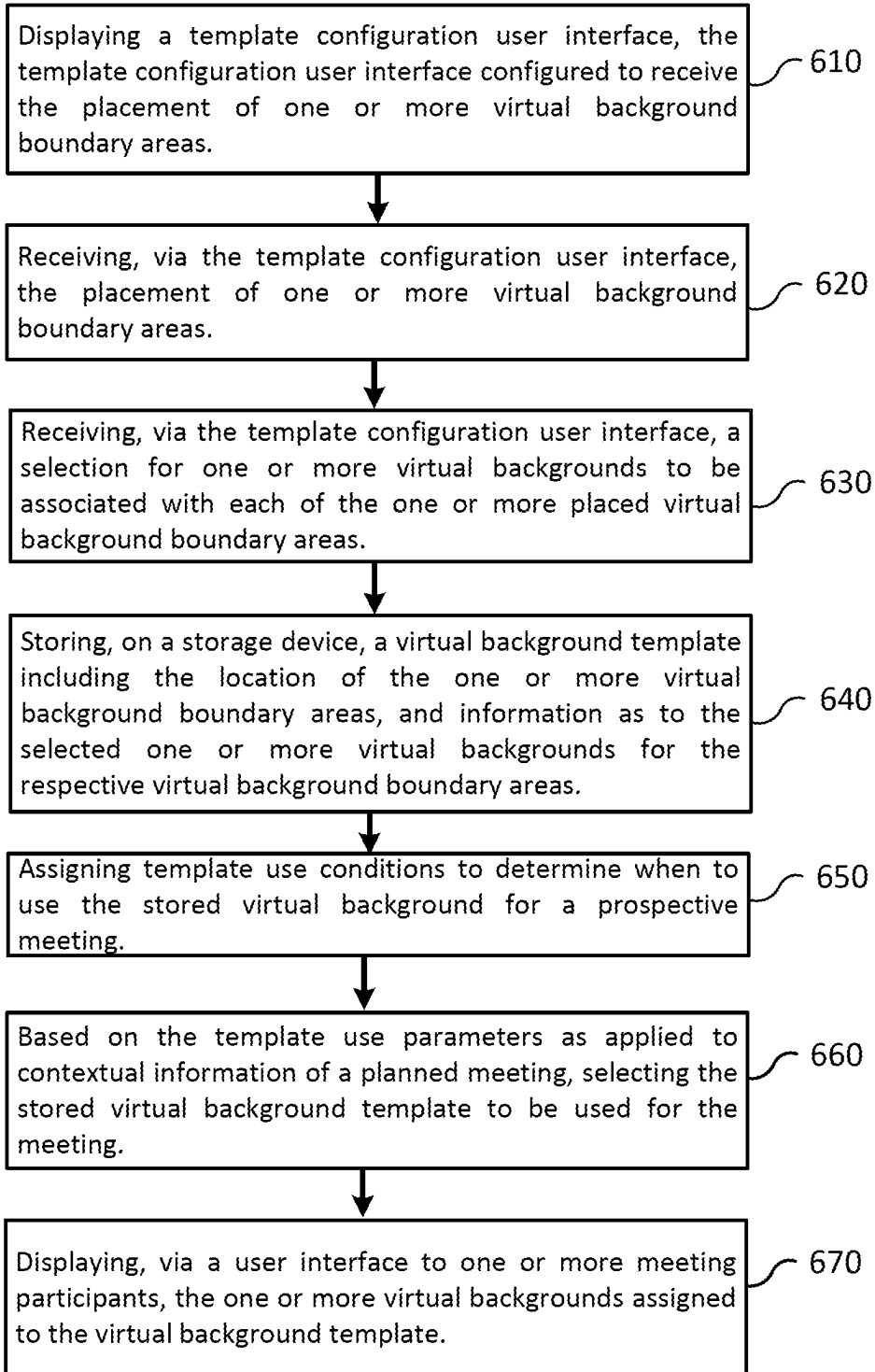


Figure 6

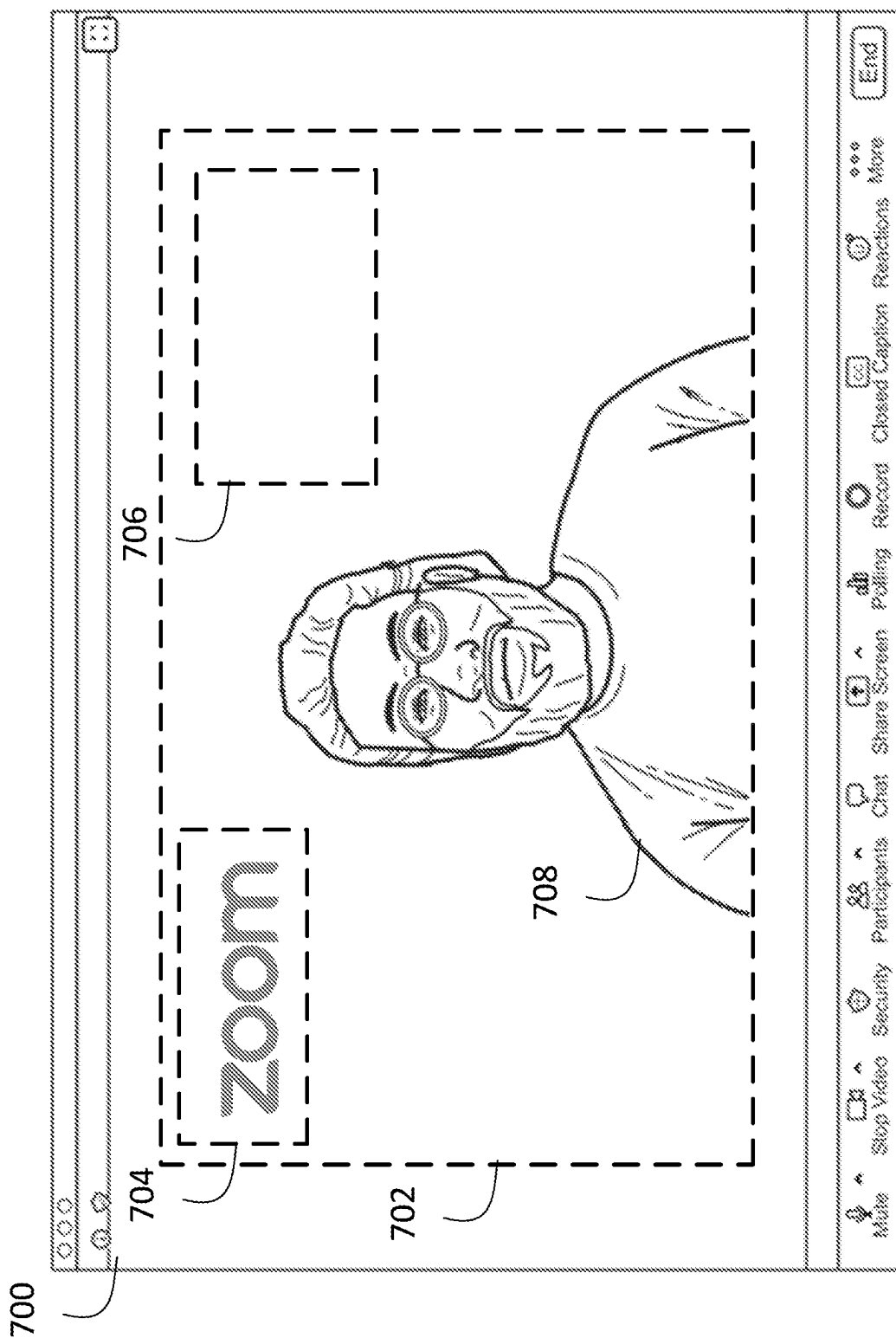
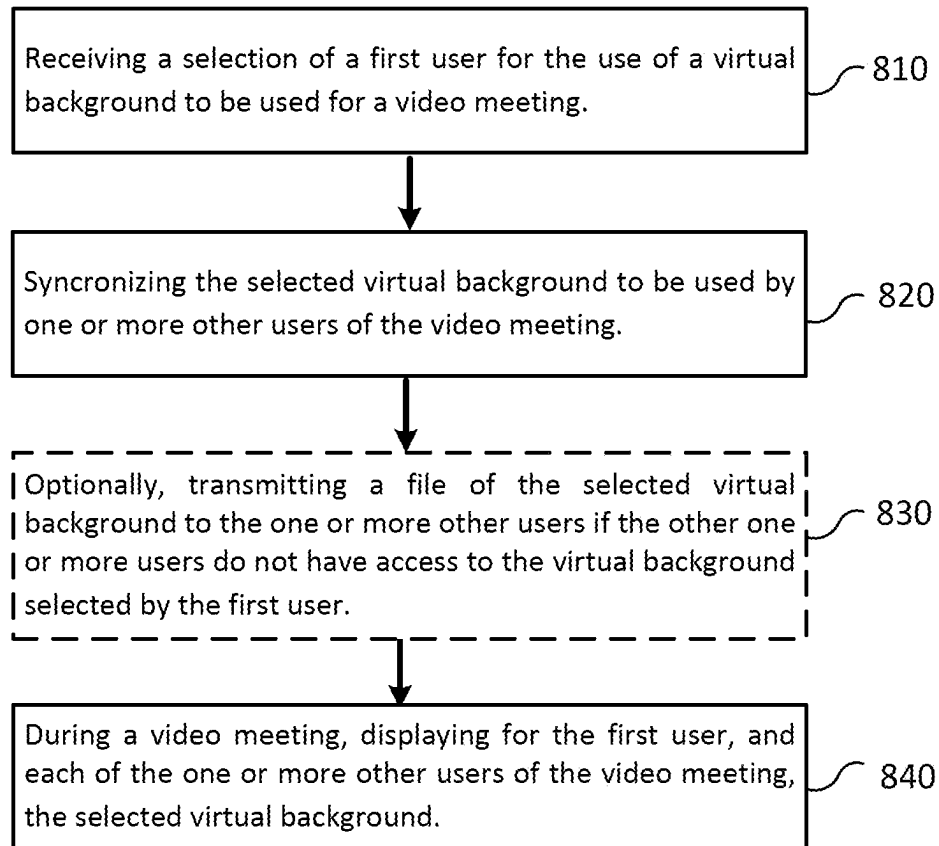


Figure 7

Virtual Background Synchronizing800**Figure 8**

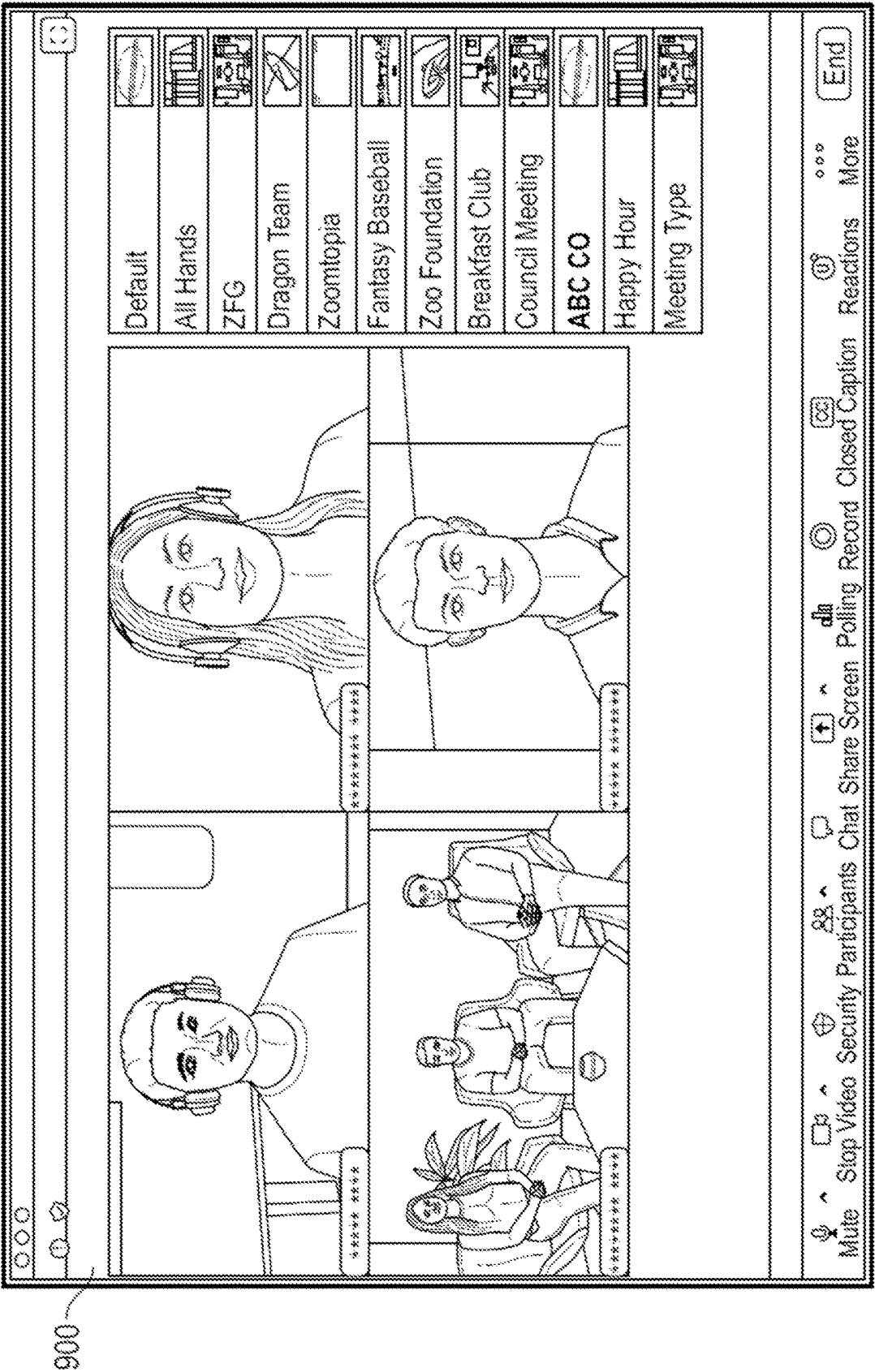
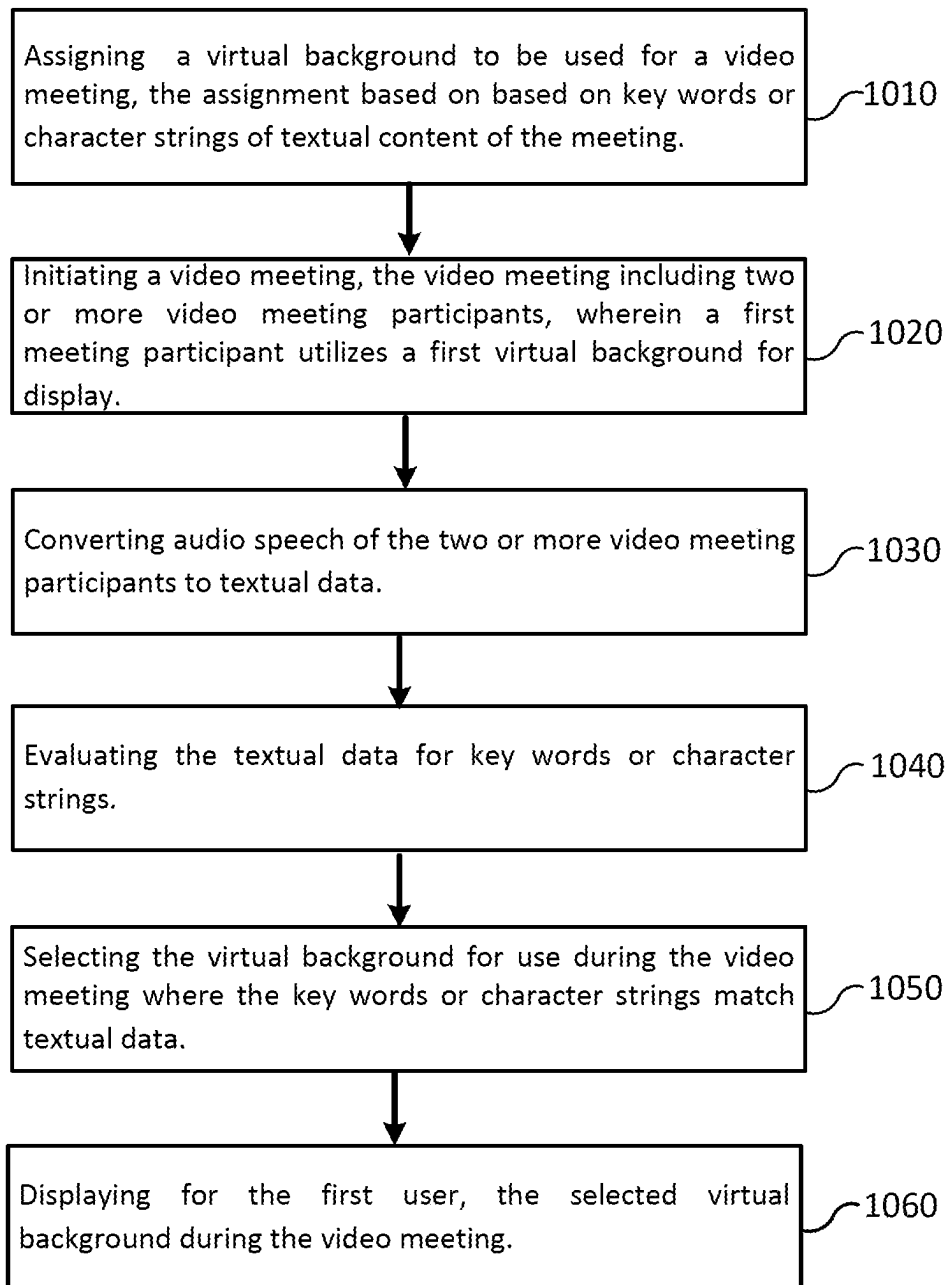


Figure 9

Dynamic Background Selection1000**Figure 10**

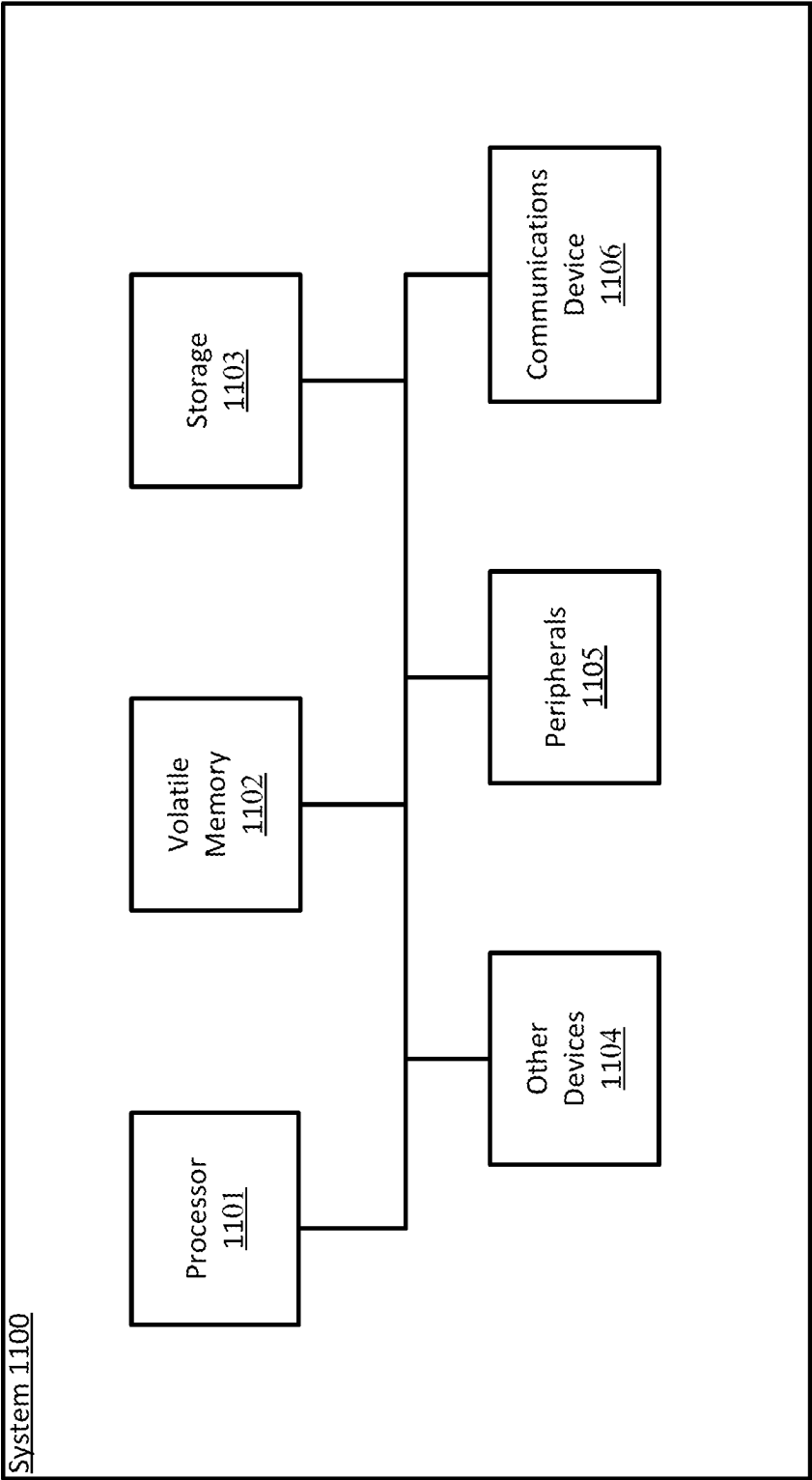


Figure 11

VIRTUAL BACKGROUND SELECTION BASED ON COMMON MEETING DETAILS

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/246,570, filed Apr. 30, 2021, the entire disclosure of which is incorporated herein by reference.

FIELD OF INVENTION

The present invention relates generally to digital communication, and more particularly, to systems and methods providing for video communications with dynamic virtual backgrounds within a video communication platform.

BACKGROUND

During video communications while using a video conferencing application or system, such as on a mobile device, smart phone or a laptop, a user may select a virtual background that is displayed as an image or video behind a live video feed of a user. These systems may use computer vision processing techniques to extract an outline (e.g., a border of the user) and then display the virtual background about or around the user during video communications with other users. However, users need to manually select a virtual background to be used for a meeting or assign a default virtual background to be used for all meetings. While providing some flexibility in the choice of a virtual background to use for video communication, contextual information of a planned meeting with other users is not utilized by these systems. This often results in situations where users may have inconsistent virtual backgrounds, for example, when employees of one company meet with employees of another company, some of the employees of the same company may be using different virtual backgrounds. The current state-of-the-art video conferencing systems lack the ability to automatically provide for consistency and uniformity of the virtual backgrounds used among users participating in a video meeting. These current state-of-the-art systems also lack the ability to automatically or dynamically choose a virtual background to be used by meeting participants.

Thus, there is a need in the field of digital communication to create a new and useful system and method for providing video communications with dynamic virtual background selection within a video communication platform. The source of the problem, as discovered by the inventors, is a lack of an ability of video communication systems to automatically generate rules for the selection of virtual backgrounds, where the selection rules are based on previous meeting information from a calendaring system.

SUMMARY

The invention overcomes the existing problems by providing a system that generates rules for the automatic selection and use of virtual backgrounds for use during a video meeting within a video communication platform. The system leverages an integrated calendar connection to maintain awareness of various user calendar details. These calendar details are evaluated by the system to automatically assign a particular virtual background for a user based upon predefined virtual background usage rules.

The video communication system provides functionality to automatically create rules where the system selects a virtual background to be used by the user during video communications with other users of the system. The system generated rules provide for the automatic selection of virtual background based on contextual information of a planned meeting. The rules are evaluated by the system and compared against information of the planned meeting. Based on information of the planned meeting, the system may determine a virtual background to be used by the user for the planned meeting.

The system defined rules may include key words and Boolean logical operators to be compared against the information of the planned meeting. The rules may be simple comparisons of key words to be found in a planned meeting, or complex rules using various syntax and logical constructs to determine whether the rule criteria or conditions are met.

The system is flexible in the sources of where meeting information are retrieved, and the manner in which a virtual background may be selected for use. The planned meeting information may be retrieved from a calendaring application or other databases that stores electronic meeting information. These data sources may be resident on a user device and/or stored in a cloud-based storage system or service.

The rules may evaluate various data of the planned meeting information, such as whether particular users or participants are attending the planned meeting, whether particular e-mail domain names exist, whether certain words occur in information descriptive of meeting, whether the meeting is scheduled during certain times or dates, the number of meeting participants, and so forth.

If the rule criteria or conditions are met, then the system selects a virtual background that has been associated with the rule. The system, for example, may generate a rule and then the system may select or assign a virtual background to be used when the system determines that the rule criteria is met.

The system may generate rules based on an evaluation or comparison of historic meetings and identifying a particular virtual background that was used for multiple meetings. For example, the system may identify that the same virtual background was used by a user for multiple meetings. The system tries to identify commonality of information of the multiple meetings, such as key words in the title, the user names and/or email addresses of the users, domain names of the e-mail addresses of the meeting participants, and so forth.

The video communications system may be placed in a mode selectable by a user to “find” or automatically suggest rules for use cases. In this discovery mode, the system may identify potential rules and present the rules and a graphical representation of a virtual background that would be used for the rule. The user interface may receive a selection from the user confirming to use the suggested rule and the associated virtual background. After receiving confirmation to use the rule, the system then would apply the system generated rule and select the associated virtual background when the rule criteria is met.

Methods, systems, and apparatus, including computer programs encoded on computer storage media relate to a method for providing video communications with dynamic virtual backgrounds within a communication platform. The system establishes a connection with a calendar account of a user. The calendar account maintains historic and planned meeting details of the user. The system may evaluate the calendar account and automatically generate rules for determining which virtual backgrounds to be used for planned

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meetings. Based on contextual information of a planned meeting, the system may apply one of the generated rules and select a virtual background to be used for video communications during the meeting.

One embodiment relates to a communication system configured to perform a number of operations. First, the system establishes a connection to a calendar account of a first user. The calendar account is capable of storing informational details about one or more meetings. For example, the informational details may include a meeting title, meeting participant e-mail addresses, a meeting date and time, a meeting location, and meeting description details. The system evaluates the informational details of prior meetings of the calendar account of the first user. Based on an evaluation of the informational details of the prior meetings of the first user, the system generates one or more rules for the selection of a virtual background to use for planned meetings by the first user. The system determines by application of the generated one or more rules, a virtual background to be used for a planned meeting.

Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for illustration only and are not intended to limit the scope of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The present invention relates generally to digital communication, and more particularly, to systems and methods providing for containment of sensitive data within a communication or messaging platform.

The present disclosure will become better understood from the detailed description and the drawings, wherein:

FIG. 1A is a diagram illustrating an exemplary environment in which some embodiments may operate.

FIG. 1B is a diagram illustrating an exemplary computer system that may execute instructions to perform some of the methods herein.

FIG. 2 is a diagram illustrating an example of planned meeting contextual information.

FIG. 3 is a flow chart illustrating an exemplary method 300 that may be performed in some embodiments.

FIG. 4 is a flow chart illustrating an exemplary method 400 that may be performed in some embodiments.

FIG. 5A illustrates an example user interface 500 according to one embodiment of the present disclosure.

FIG. 5B illustrates an example user interface 500 according to one embodiment of the present disclosure.

FIG. 5C illustrates an example user interface 500 according to one embodiment of the present disclosure.

FIG. 6 is a flow chart illustrating an exemplary method 600 that may be performed in some embodiments.

FIG. 7 illustrates an example user interface 700 according to one embodiment of the present disclosure.

FIG. 8 is a flow chart illustrating an exemplary method 800 that may be performed in some embodiments.

FIG. 9 illustrates an example user interface 900 according to one embodiment of the present disclosure.

FIG. 10 is a flow chart illustrating an exemplary method 1000 that may be performed in some embodiments.

FIG. 11 is a diagram illustrating an exemplary computer system 1100 that may perform processing in some embodiments.

DETAILED DESCRIPTION

In this specification, reference is made in detail to specific embodiments of the invention. Some of the embodiments or their aspects are illustrated in the drawings.

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For clarity in explanation, the invention has been described with reference to specific embodiments, however it should be understood that the invention is not limited to the described embodiments. On the contrary, the invention covers alternatives, modifications, and equivalents as may be included within its scope as defined by any patent claims. The following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations on, the claimed invention. In the following description, specific details are set forth in order to provide a thorough understanding of the present invention. The present invention may be practiced without some or all of these specific details. In addition, well known features may not have been described in detail to avoid unnecessarily obscuring the invention.

In addition, it should be understood that steps of the exemplary methods set forth in this exemplary patent can be performed in different orders than the order presented in this specification. Furthermore, some steps of the exemplary methods may be performed in parallel rather than being performed sequentially. Also, the steps of the exemplary methods may be performed in a network environment in which some steps are performed by different computers in the networked environment.

Some embodiments are implemented by a computer system. A computer system may include a processor, a memory, and a non-transitory computer-readable medium. The memory and non-transitory medium may store instructions for performing methods and steps described herein.

Many other possibilities and options can be contemplated for this use case and others, as will be described in further detail throughout.

I. Exemplary Environments

FIG. 1A is a diagram illustrating an exemplary environment in which some embodiments may operate. In the exemplary environment 100, a first user's client device 150 and one or more additional users' client device(s) 160 are connected to a processing engine 102 and, optionally, a video communication platform 140. The processing engine 102 is connected to the video communication platform 140, and optionally connected to one or more repositories and/or databases, including a user account repository 130, video content repository 132, and/or an annotation repository 134. One or more of the databases may be combined or split into multiple databases. The first user's client device 150 and additional users' client device(s) 160 in this environment may be computers, and the video communication platform server 140 and processing engine 102 may be applications or software hosted on a computer or multiple computers which are communicatively coupled via remote server or locally.

The exemplary environment 100 is illustrated with only one additional user's client device, one processing engine, and one video communication platform, though in practice there may be more or fewer additional users' client devices, processing engines, and/or video communication platforms. In some embodiments, one or more of the first user's client device, additional users' client devices, processing engine, and/or video communication platform may be part of the same computer or device.

In an embodiment, the processing engine 102 may perform the method 200 (FIG. 2A) or other method herein and, as a result, provide multi-point video presentations with live annotations within a video communication platform. In some embodiments, this may be accomplished via communication with the first user's client device, additional users'

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client device(s), processing engine, video communication platform, and/or other device(s) over a network between the device(s) and an application server or some other network server. In some embodiments, the processing engine **102** is an application, browser extension, or other piece of software hosted on a computer or similar device, or is itself a computer or similar device configured to host an application, browser extension, or other piece of software to perform some of the methods and embodiments herein.

The first user's client device **150** and additional users' client device(s) **160** are devices with a display configured to present information to a user of the device. In some embodiments, the first user's client device **150** and additional users' client device(s) **160** present information in the form of a user interface (UI) with UI elements or components. In some embodiments, the first user's client device **150** and additional users' client device(s) **160** send and receive signals and/or information to the processing engine **102** and/or video communication platform **140**. The first user's client device **150** is configured to perform functions related to presenting and playing back video, audio, documents, annotations, and other materials within a video presentation (e.g., a virtual class, lecture, webinar, or any other suitable video presentation) on a video communication platform. The additional users' client device(s) **160** are configured to viewing the video presentation, and in some cases, presenting material and/or video as well. In some embodiments, first user's client device **150** and/or additional users' client device(s) **160** include an embedded or connected camera which is capable of generating and transmitting video content in real time or substantially real time. For example, one or more of the client devices may be smartphones with built-in cameras, and the smartphone operating software or applications may provide the ability to broadcast live streams based on the video generated by the built-in cameras. In some embodiments, the first user's client device **150** and additional users' client device(s) are computing devices capable of hosting and executing one or more applications or other programs capable of sending and/or receiving information. In some embodiments, the first user's client device **150** and/or additional users' client device(s) **160** may be a computer desktop or laptop, mobile phone, virtual assistant, virtual reality or augmented reality device, wearable, or any other suitable device capable of sending and receiving information. In some embodiments, the processing engine **102** and/or video communication platform **140** may be hosted in whole or in part as an application or web service executed on the first user's client device **150** and/or additional users' client device(s) **160**. In some embodiments, one or more of the video communication platform **140**, processing engine **102**, and first user's client device **150** or additional users' client devices **160** may be the same device. In some embodiments, the first user's client device **150** is associated with a first user account on the video communication platform, and the additional users' client device(s) **160** are associated with additional user account(s) on the video communication platform.

In some embodiments, optional repositories can include one or more of a user account repository **130**, meetings repository **132**, virtual background repository **134** and/or virtual background configuration repository **136**. The optional repositories function to store and/or maintain, respectively, user account information associated with the video communication platform **140**, video content received by the video communication platform **140** for display within video communication, and virtual background, which may include images, video and other graphical visual content.

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The optional database(s) may also store and/or maintain any other suitable information for the processing engine **102** or video communication platform **140** to perform elements of the methods and systems herein. In some embodiments, the optional database(s) can be queried by one or more components of system **100** (e.g., by the processing engine **102**), and specific stored data in the database(s) can be retrieved.

Video communication platform **140** is a platform configured to facilitate video presentations and/or communication between two or more parties, such as within a video conference or virtual classroom.

FIG. **1B** is a diagram illustrating an exemplary computer system **150** with software modules that may execute some of the functionality described herein.

The Planned Meeting Evaluation module **152** provides system functionality for the interaction and retrieval planned meeting contextual information from calendaring systems and/or database, and as further described herein.

The Virtual Background Template Configuration module **154** provides system functionality for the configuration, processing, storage and management of virtual background templates, and as further described herein.

The Virtual Background Rules Configuration module **156** provides system functionality for the selection, creation, configuration, processing, storage and management of virtual background selection rules, and as further described herein.

Virtual Background Selection module **158** provides system functionality for the selection, creation, configuration, processing, storage and management of particular virtual backgrounds.

Virtual Real-time Meeting Monitoring module **160** provides system functionality for the monitoring and evaluation of meeting that are in progress to identify and determine real-time video communications, audio communications and textual communications, and as further described herein.

Virtual Background Synchronization module **162** provides system functionality for the synchronization of virtual backgrounds among users of a meeting, and as further described herein.

The above modules and their functions will be described in further detail in relation to an exemplary methods and systems below.

Virtual Background Files

A virtual background file may be a file such as a video file, an image files (e.g., a jpeg, gif, etc.), or any other type of graphics or multimedia file. In general, a virtual background file is a file of any type that allows for the system to present a still graphic image and/or a video image as a virtual background in conjunction with a user's video feed. The virtual background file may be stored on a file system, computer system memory, either in local storage or in a server-based storage system or database. The system retrieves the virtual background file to be used by the system as a virtual background while a user is engaged in video communications with one or more other users. The virtual background files may be cataloged or indexed by the system and associated with rules for the selection of particular virtual background files.

Planned Meeting Contextual Information

FIG. **2** is a diagram illustrating an example of planned meeting contextual information. The system **100** may retrieve from a meetings repository, such as a calendar application, information that is specific to a planned meeting event. By way of illustration, FIG. **2** depicts common information typically stored by a calendaring application. A planned meeting event may include a title **202**, a meeting

location **204**, a meeting date and time **206**, a list of meeting attendees **210** and other information. The listing of meeting attendees usually are identified in the format of username@domain-name. For example, a meeting attending might be identified with the email address JAppleseed@zoom.com. Often a user creating a meeting, via a calendaring application, will provide a meeting title **202** to provide context for the meeting.

In some embodiments, preconfigured rules (as described herein) may use conditions and/or parameters that are evaluated against contextual information of a planned meeting. For instance, by way of illustration, but not limitation, some contextual information that may be evaluated by the system include: the title of the meeting, number of meeting participants, the time zone of one or more meeting participants, the description of the meeting, the particular participant user ids, email address or time zone of meeting participants.

The planned meeting contextual information may be presented, via a user interface, such as a calendaring application user interface. In some embodiments, the calendaring application user interface may provide for the association of a virtual background file, via user interaction with the calendaring application user interface. For example, the calendaring application user may display a graphical icon where a user may upload or assign a particular virtual background to be used by one or more attendees of a meeting.

Virtual Background Selection Rules

In some embodiments, preconfigured rules for virtual background selection may be user defined, automatically generated by the system, preconfigured by the system for a user and/or the rules shared among users. The system provides functionality allowing a user to define conditions and/or parameters where a virtual background file is automatically selected for use as a virtual background for prospective meetings. In determining which virtual background to select, the system evaluates contextual information of a planned meeting, and then applies the preconfigured rule to the contextual information to determine whether the criteria for the selection of a virtual background are met. Where the conditions and/or parameters for the selection of a virtual background are met, then the system selects the virtual background to be used for user video communications. Accordingly, the virtual background is selected automatically, without requiring the user confirming the selection.

The system may automatically generate preconfigured rules for a user for the selection of a virtual background to be used for video communications. The system may evaluate historical meeting information from a calendaring system, or other electronic repository having past meeting information for the user. The system may store information about a prior selected virtual background that was used or assigned for use for users for various meetings. The system may determine commonality about contextual information of the various meeting, and automatically generate a preconfigured rule for a user to select a virtual background for prospective meetings. For example, the system may determine that a user used the same virtual background where the meeting participants included a specific user email domain name, or a particular user email address. The system then may create a preconfigured rule to automatically select the same virtual background that was previously used, where the user has a planned meeting with any other users having a similar email domain name and/or if the particular user email address is found in the contextual information of a planned meeting.

In another example, the system may determine that a user typically uses the same virtual background (such as an image with the user's company logo) where the title of a meeting (as found in the meeting contextual information) is called "Company All-hands". The system may then generate a preconfigured rule for that user where the system would evaluate the contextual information of a planned meeting for keywords of "Company All-hands" (either case-sensitive or not), and then associate the previously used virtual background for the planned meeting where the preconfigured rule criteria is met. In this instance, the system would automatically select the image of the user's company logo, when the title of a planned meeting includes the keywords "Company All-hands".

The system provides functionality where users may share their preconfigured rules with other users. The system may provide a user interface where a user may select one or more preconfigured rules to send to one or more other users. The selection user interface may allow for the input of user names and/or email addresses to whom the preconfigured rules would be shared. When sharing the preconfigured rule, the system provides the rule name, conditions and parameters, and copy of the underlying virtual background file to the other user. This functionality provides for one user to share a preconfigured rule with the other users. The other users may now use or modify the preconfigured rules for their own meetings.

In some embodiments, the preconfigured rules may be given a rule priority order or weight for consideration by the system. In other words, the system may receive an indication that a rule has a higher priority than another rule. For example, the preconfigured rules may be given a rank or priority order for evaluation by the system. In one embodiment, a user interface may display a listing of the preconfigured rules, and the order of the rules (for example, top to bottom) indicate the order or priority that the system will evaluate the rule. In another embodiment, a user interface may receive a ranking number to indicate the order the preconfigured rule would be evaluated by the system.

In some embodiments, the conditions and/or parameters of a preconfigured rule may include key words to match, Boolean logic operators to find matches (such as "and", "or"), wild card operators and different character strings. A preconfigured rule may also include negative operators to preclude when a virtual background is not to be used. For example, a user may have personal based virtual backgrounds that are used when video conferencing with family or friends, and may not want to have the virtual background used for company or business meetings. The user may identify conditions and/or parameters of when not to use a particular virtual background for certain meeting contextual information. In this instance, this would preclude use of the virtual background from being automatically selected for those do not use meeting situations. Also, the system may be configured to preclude the user from manually selecting the virtual background in such situations, or at least prompt the user noting the virtual background has been precluded for use for the particular meeting at hand, and then allowing the user to override the preconfigured rule is the user so desires.

In some embodiments, the system may in some instances, select two or more virtual backgrounds that may be used for a video meeting based on the application of two or more preconfigured rules as applied to contextual information of a planned meeting. In such a case, the system may present to the user a user interface depicting two or more graphical representations of the selected two or more virtual backgrounds. The system may receive a selection from a user for

one of the displayed graphical representations. The system then would use the virtual background associated with the received selection of the graphical representation for use as a virtual background for the user during video communications.

II. Exemplary Methods and User Interfaces

Virtual Background Selection

FIG. 3 is a flow chart illustrating an exemplary method 300 that may be performed in some embodiments. In one embodiment, the system 100 provides for virtual background selection using preconfigured rules. The system 100 provides functionality where a user may define a rule to select a particular virtual background to be used for a planned video meeting. The system 100 evaluates planned meeting contextual information and determines a specific virtual background to be used for video communication. In some instances, two or more rules may be applicable and result in the selection of two different virtual backgrounds for use. In the case, the system may present a user interface displaying a representation of the two virtual backgrounds, and receive a selection for the desired virtual background to be used during video communications by the user.

At step 310, the system 100 provides one or more preconfigured rules for the selection of one or more virtual backgrounds to be used for planned video meetings. The preconfigured rules may be stored on a local or cloud data storage device and retrieved by the system.

At step 320, the system 100 receives contextual information about a planned meeting. For example, the system may retrieve contextual information from a calendaring system or other database storing contextual information about planned meetings.

At step 330, the system 100 selects at least one virtual background for use by a first user for the planned meeting. The system may select a virtual background based on the application of one or more preconfigured rules as applied to the received contextual information of the planned meeting.

At step 340, the system 100 receives video content depicting imagery of the first user, the video content having multiple video frames. The system provides functionality for a user to capture and display video imagery to other users. For example, the system may receive a video stream from digital camera depicting imagery of the first user.

At step 350, the system determines a boundary about the first user in the video frames, wherein the boundary has an interior portion and exterior portion. The system may use various computer vision techniques to identify the border or outline of the user depicted in the video.

At step 360, the system 100 generates, for display on one or more client devices, a composite video depicting the interior portion depicting imagery of the first user overlaid on imagery of the selected virtual background. Thus, the user imagery is added to (i.e., composited with) the selected virtual background. The system combines or composites video imagery of the user shown with the selected virtual background.

FIG. 4 is a flow chart illustrating an exemplary method 400 that may be performed in some embodiments. The method 400 show an example of processing multiple preconfigured rules, where the rules are evaluating the meeting contextual information.

The method of evaluating meeting contextual information is illustrated for a search and match process for multiple preconfigured rules, with each preconfigured rule searching for and matching key words. The system may retrieve from

a data storage device multiple preconfigured to determine a virtual background filed to be selected for a user as a virtual background during video communications with other users. In step 410, the system evaluates one or more retrieved n preconfigured rules. In step 420, the system determines particular keys words for the particular nth rule. The system determines the rule's logical operators and syntax, for example, determining which key words to match to planned meeting contextual information. For example, one or more key words may be required to be matched to meeting contextual information for a particular virtual background file to be selected.

At step 430, the system compares or tries to find matches of the key words with contextual information of a planned meeting. At step 440, the system determines whether or not a match was found. For each of the key words, the system determines whether all of the key words are found. If all of the key words are matched to contextual information of the meeting, then the system may then select a virtual background file that was associated with the preconfigured rule (step 450). The system would use the virtual background file to provide the virtual background during video communications by the user. While a simple key word match may be configured as a rule, more complex rules, operations, parameters and conditions may be required for a preconfigured rule. If no required key word matches are found, then next preconfigured rule is then evaluated against the contextual information, and so on until no more rules exist (looping back to step 430). The system may be configured such that a particular virtual background file would be used if none of the preconfigured rules apply to the contextual meeting information.

FIG. 5A illustrates an example user interface according to one embodiment of the present disclosure. The system 100 may generate and display a user interface 500 where a user may select a virtual background to use as a default virtual background. The user interface 500 may provide a selection for virtual background configuration where the user interface 500 provides functionality where a user may configure one or more rules for the selection of a virtual background to be used for a planned video meeting.

FIG. 5B illustrates an example user interface 500 according to one embodiment of the present disclosure. In response to the selection of "Virtual Background Configuration" text, the system 100 may display the user interface section 530. The user interface section 530 depicts a listing of key terms and a group of virtual background from which a virtual background may be selected to be used when the key words are found in the contextual information for a planned meeting. The listing of key words 532 may be removed by the selection of the trashcan icon to the left of a particular key word.

The following example illustrates how a user may interact with the user interface section 530. A user may want to set or select a virtual background to be used for planned meeting. The user would enter key words into the input text box 534. In the examples 532A, 532B and 532C, a user had typed in text in the user interface input text box 534. From the listing of virtual backgrounds, the user had also selected a particular virtual background to be used for the input key words.

In one example, for the key words "ABC CO" 532A, a user may have selected the graphical representation of virtual background 542A to be used when a planned meeting's contextual information includes the key words "ABC CO". As will be described further below, the system 100 may evaluate the planned meeting contextual information to

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identify that the planned meeting has the phrase “ABC CO”. In such a case, the system **100** will select and utilize the virtual background that is associated with the graphical representation of virtual background **542A** to be used as a virtual background for the user during video communications with other users.

In another example, for the key words “Weekly Review” **532B**, a user may have selected the graphical representation of virtual background **542B** to be used when a planned meeting’s contextual information includes the key words “Weekly Review”. As will be described further below, the system **100** may evaluate the planned meeting contextual information to identify that the planned meeting has the phrase “Weekly Review” for instance in the title of the meeting. In such a case, the system **100** will select and utilize the virtual background that is associated with the graphical representation of virtual background **542B** to be used as a virtual background for the user during video communications with other users.

In yet another example, for the key words “@zoom.com” **532C**, a user may have selected the graphical representation of virtual background **542C** to be used when a planned meeting’s contextual information includes the key words “@zoom.com”. As will be described further below, the system **100** may evaluate the planned meeting contextual information to identify that the planned meeting has the phrase “@zoom.com”, for instance in the e-mail address of one of the meeting participants. In such a case, the system **100** will select and utilize the virtual background that is associated with the graphical representation of virtual background **542C** to be used as a virtual background for the user during video communications with other users.

FIG. **5C** illustrates an example user interface according to one embodiment of the present disclosure. In this embodiment of the user interface **500**, the system **100** may present the user interface with an input section where the user interface **500** may receive a selection for the name of a rule, a selection or input for a user calendar associated with the rule (e.g., Outlook.com, Gmail.com, iCloud.com, etc.), and one or more key words to be matched for a planned meeting’s contextual information. In this embodiment, the user interface may upload a virtual background to the system local or cloud-based storage repository in association with an account and/or user id of the user.

Automatic Virtual Background Rule Generation

In some embodiments, the system **100** may evaluate the historical use of virtual backgrounds that were previously used for a meeting, and automatically construct rules for use of virtual backgrounds for planned meetings. As described previously, the system may determine that a previous virtual background was used for many different meetings. The system may perform processing to determine similarities of contextual information of the different meetings and derive or create rules to use the same virtual background where a planned meeting has similar contextual information as the historical meetings.

In this embodiment, the system may establish a connection to the calendar account of a first user, or other database storing previous meeting information. A calendar account of a user is capable of storing informational details about one or more meetings. The informational details, for example, may include for each meeting: a title for the meeting, participant e-mail addresses of the meeting, a date for the meeting and a time for the meeting. Also, the system **100** may store information for particular virtual backgrounds that were used for past meetings.

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The system **100** may evaluate the informational details of prior meetings of the calendar account of the first user. The system may retrieve and generate a dataset including the information details for those meetings where the same virtual background was used.

The system **100** may generate one or more rules for the selection of a virtual background to be used for prospective meetings by the first user. The generated one or more rules may be generated based on an evaluation of the informational details of the prior meetings. For example, the system may determine that similar key words exist in the contextual information for the meetings. The system may then generate a rule where the system would use the same virtual background as previously used, where the key words are found in the contextual information for a planned meeting.

The system **100** may determine, by applying the generated one or more rules, a virtual background to use for a planned meeting.

Virtual Background Template Configuration

FIG. **6** is a flow chart illustrating an exemplary method **600** that may be performed in some embodiments. The system **100** provides functionality to display a user interface (as further described with respect to FIG. **7**) where a user may define a template for the selection of one or more virtual backgrounds to be used for a planned video meeting. The system may then evaluate planned meeting contextual information and select a specific virtual background template to be used for video communication. During video communications by the user, the system would use the virtual backgrounds configured for the selected virtual background template.

At step **610**, the system displays a template configuration user interface. The template configuration user interface may be configured to receive the placement of one or more virtual background boundary areas.

At step **620**, the system receives, via the template configuration user interface, the placement of one or more virtual background boundary areas. The user interface may receive selection about the template configuration interface of where the user would like to add a virtual background boundary area.

At step **630**, the system receives, via the template configuration user interface, a selection for one or more virtual backgrounds to be associated with each of the one or more placed virtual background boundary areas. For each of the virtual background boundary areas, a user may select a virtual background to be displayed within a particular virtual background boundary area.

At step **640**, the system stores, on a storage device, the user configured virtual background template including the location of the one or more virtual background boundary areas about the user interface, and the information as to the selected one or more virtual backgrounds for the respective virtual background boundary areas.

At step **650**, the system, assigns use conditions/parameters to determine when to use the stored virtual background template for a planned meeting. The use conditions may be preconfigured rules as described herein, such as assigning keywords to be matched as to the contextual information of a planned meeting.

At step **660**, the system based on the template use conditions/parameters as applied to the contextual information of a planned meeting, selecting the stored virtual background template to be used for the meeting.

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At step **670**, the system **100** displays, via a user interface, to one or more meeting participants during a video meeting, the one or more virtual backgrounds assigned to the virtual background template.

Virtual Background Template

FIG. 7 illustrates an example user interface **700** according to one embodiment of the present disclosure. The system **100** may generate and display a user interface **700** where a user may configure a virtual background template to use for video communications with other users. The user interface **700** may provide functionality for the selection and placement of sections and/or borders (i.e., virtual background boundary areas) on the user interface **700** for the display of one or more virtual backgrounds. The virtual background template during configuration may optionally display a video feed of a user **708** during the configuration of the virtual background template.

The system may generate and display the user interface **700** which allows a user to design or configure one or more virtual backgrounds to be displayed during a video communication with other users. A user may select, draw, move, delete and/or resize one or more virtual background boundary areas on the user interface **700** (e.g., using a selection tool via a mouse input, touch input or other input device). The user interface **700** may receive the placement of one or more virtual background boundary area onto the user interface. For example, the user interface **700** depicts three virtual background boundary areas showing a desired placement of three different virtual background boundary areas (represented by the dashed borders **702**, **704**, **706**). A virtual background boundary area may be layered such that top layer (or most forward positioned) virtual background boundary area displays a virtual background over another virtual background during the video communications.

A virtual background boundary area may be associated with a particular virtual background to be displayed during video communications when the virtual background template is selected for use for a meeting. As an example, the virtual background boundary area **704** is associated with a virtual background of a graphical image of the word “**ZOOM**”.

The associated virtual backgrounds for the respective virtual background boundary areas may be stored collectively together as a virtual background template. The virtual background template may include the locations, shape and size of the virtual background boundary areas, and an indication such as a file name, link, identifier or other reference for a specific virtual background to be used in association with the virtual background boundary area. The virtual background template may be stored on a local data storage device and/or a remote storage service. In some embodiments, the virtual background files may be stored along with the virtual background as a collective container of the data, information and files needed to use or recreate the virtual background template.

A particular virtual background template may be associated with pre-configuration rules (i.e., similar to the virtual background rule configuration as discussed previously). Additionally, each of the individual virtual background boundary areas may be associated with the same or different pre-configuration rules. A virtual background boundary area may be associated with a rule that if key words are matched in the contextual information of a planned meeting, then the virtual background boundary area would be displayed. For example, if a preconfigured rule includes the keywords “zoom.com”, and the contextual information includes a reference to zoom.com, such as a user email address, then

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the virtual background boundary area **704** would be initiated and become active during the video meeting, and the system would display the associated virtual background (e.g., the graphical image of the word “**ZOOM**”).

In some embodiments, each of the virtual background boundary areas may be associated with different or similar preconfigured rules. Also, a particular virtual background boundary area may be associated with multiple preconfigured rules. For example, the virtual background boundary area **704** may also be associated with the keywords “Baseball”, and virtual background (such as a logo of one’s favorite baseball team) may be associated with the keywords and the virtual background boundary area **704**. In this instance, if a planned meeting includes contextual information include a keyword for baseball, then the system **100** would present associated virtual background for the associated baseball team logo. This functionality allows a user to configure a dynamic virtual background template that displays different virtual backgrounds for the same virtual background boundary areas based on a preconfigured rule and the informational content of a planned meeting.

In some embodiments, the user interface **700** may receive a textual input directly into the virtual background boundary area. For example, the user interface **700** may receive the input for the word Hobby, CompanyName, etc. The system **100** may receive and store user input tags for different virtual backgrounds. For example, system **100** may receive a tag from a user input, associating a particular virtual background with the tag. In one instance an image to be used as a virtual background may be tagged with the word “Hobby” and another image tagged with a company name. Then for a planned video meeting, the virtual background would be used where the word “Hobby” or the company name is found in the contextual information for the meeting.

Additionally, the virtual background boundary area may be set as random. In this case, a different virtual background may be automatically chosen by the system **100** for each new video meeting in which the user participates.

Virtual Background Synchronization

FIG. 8 is a flow chart illustrating an exemplary method **800** that may be performed in some embodiments. The system **100** provides functionality to display a user interface where a first user may select a virtual background to be used for a video meeting. The system **100** will synchronize the virtual background of the other user with the virtual background selected by the first user. The system **100** may optionally transmit a file for the virtual background to those users that do not already have access to the same virtual background selected by the first user. During a video meeting the selected virtual background would be used by the first user and at least one other user.

At step **810**, the system receives a selection of a first user for the use of a virtual background to be used for a video meeting. For example, a user may select which virtual background should be used for a meeting. Alternatively, the system may use a preconfigured rule to select a virtual background to be used for the meeting.

At step **820**, the system synchronizes the selected virtual background to be used by one or more other users of the video meeting. The system provides for the selection of one or more other user to use the selected virtual background. This functionality provides, for example, a meeting admin, meeting host, etc., to enforce a common virtual background to be used by the other meeting participants.

At step **830**, the system optionally transmits a file of the selected virtual background to the one or more other users if the other one or more users do not have access to the virtual

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background selected by the first users. For example, a user may not have access to virtual background selected by the first user. In this instance, the system may transmit a file of the selected virtual background to the other user. The other user may then receive and store the received virtual background on their local or cloud data storage device.

At step 840, during a video meeting, the system displays for the first user, and each of the one or more other users of the video meeting, the selected virtual background. The first user and the other users would display during the video meeting the virtual background as selected by the first user.

FIG. 9 illustrates an example user interface according to one embodiment of the present disclosure. In some embodiments, the user interface allows the presentation of different virtual backgrounds for selection by a user. The system retrieves from a data storage device one or more virtual background files that correspond to preconfigured rules for the user. The displayed listing of virtual backgrounds for one user may be different than the listing of virtual backgrounds for other users. The selection and display of the virtual backgrounds may depend upon the preconfigured rules for each user and on the contextual information of the meeting in which the users participate.

In some embodiments, the system synchronizes the displayed virtual backgrounds and displays at least a group of similar virtual backgrounds for each user participating in the meeting. For example, the system may determine that two or more users have a similar virtual background that may be chosen based on one or more preconfigured rules for the selection of a virtual background based on the contextual information of the meeting. In such instances, where the system determines that each of the users have a same virtual background that may be used for the video meeting, the system may automatically select for use for those users the same virtual background to be used during the meeting.

In some embodiments, a user may select a virtual background to be used by other meeting participants. The system provides functionality allowing a user, such as a host or co-host of a meeting, to select one of the virtual backgrounds to be used by all of the meeting participants, a group of meeting participants, and/or specific identified meeting participants. For example, this functionality is especially useful in that the system allows a host to select which virtual background would be used for the meeting for some or all of the users. In a situation where employees of the same company are meeting with another company, a user of the company hosting the meeting can ensure a consistent virtual background to be used by all of the users of the company participating on a video conference call. Where a user selects a virtual background for use by certain users or all of users, the system would cause the selection of the particular virtual backgrounds to be used by each of the certain users. When the certain users participate in a video meeting, each of their virtual backgrounds would be uniformly displayed with the same virtual background.

In some embodiments, a user, such as a meeting host or admin, may assign a virtual background to be used by the particular user. As described with respect to the planned meeting contextual information, a primary user may assign a virtual background to be used directly for a user. During the video meeting, the primary user assigned virtual background would then be used by the respective users. In alternative embodiments, the primary user may assign a particular virtual background to be used by all meeting participants. In other words, a virtual background may be assigned at a meeting level where each meeting participant would be displayed with the same virtual background.

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Dynamic Virtual Background Selection

FIG. 10 is a flow chart illustrating an exemplary method 1000 that may be performed in some embodiments. In some embodiments, the system may evaluate meeting content and select a virtual background for a user based on the evaluated meeting context.

Speech-to-Text and Selection of Virtual Backgrounds

In one embodiment, the system may convert speech-to-text and based on an evaluation of the text determine a virtual background to use, or dynamically change to during the meeting.

In reference to FIG. 10, at step 1010, the system 100 assigns a virtual background to be used for a video meeting, the assignment based on based on key words or character strings of textual content of the meeting.

At step 1020, the system 100 initiates a video meeting, the video meeting including two or more video meeting participants, wherein a first meeting participant utilizes a first virtual background for display.

At step 1030, the system 100 convert audio speech of the two or more video meeting participants to textual data.

At step 1040, the system 100 evaluates the textual data for key words or character strings.

At step 1050, the system 100 selects the virtual background for use during the video meeting where the key words or character strings match textual data.

At step 1060, the system 100 displays for the first user, the selected virtual background during the video meeting or for at least a portion of the video meeting.

Object Detection of User Video and Selection of Virtual Backgrounds

In other embodiments, the system 100 may evaluate the video stream of a user and determine a virtual background to use for video communications based on text found in the video stream of a user. For example, the system 100 may receive video frames, evaluate the video frames and identify one or more objects in the video frames. The system 100 may determine whether an object includes textual information, such as the name on a hat and/or a logo on a t-shirt. The system 100 may then suggest to a user or automatically select a virtual background to use for video communications that is associated with the text identified in one or more objects of the video frames.

In other embodiments, the system 100 may evaluate the video stream of a user and determine a virtual background to use for video communications based on motions of the user. For example, the system 100 may receive video frames, evaluate the video frames and identify user motions in the video frames. The system 100 may determine whether a particular motion has been made (for example, hand signing or performing a gesture with the hands, or head movement). The motion may be associated with a particular virtual background to be used when the motion is identified by the system. For example, the system 100 may detect that a user is performing the motion of waving goodbye, the system 100 may in response display a virtual background associated with the detected waving goodbye motion.

Exemplary Computer System

FIG. 11 is a diagram illustrating an exemplary computer that may perform processing in some embodiments. Exemplary computer 1100 may perform operations consistent with some embodiments. The architecture of computer 1100 is exemplary. Computers can be implemented in a variety of other ways. A wide variety of computers can be used in accordance with the embodiments herein.

Processor 1101 may perform computing functions such as running computer programs. The volatile memory 1102 may

provide temporary storage of data for the processor **1101**. RAM is one kind of volatile memory. Volatile memory typically requires power to maintain its stored information. Storage **1103** provides computer storage for data, instructions, and/or arbitrary information. Non-volatile memory, which can preserve data even when not powered and including disks and flash memory, is an example of storage. Storage **1103** may be organized as a file system, database, or in other ways. Data, instructions, and information may be loaded from storage **1103** into volatile memory **1102** for processing by the processor **1101**.

The computer **1100** may include peripherals **1105**. Peripherals **1105** may include input peripherals such as a keyboard, mouse, trackball, video camera, microphone, and other input devices. Peripherals **1105** may also include output devices such as a display. Peripherals **1105** may include removable media devices such as CD-R and DVD-R recorders/players. Communications device **1106** may connect the computer **1100** to an external medium. For example, communications device **1106** may take the form of a network adapter that provides communications to a network. A computer **1100** may also include a variety of other devices **1104**. The various components of the computer **1100** may be connected by a connection medium such as a bus, crossbar, or network.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as "identifying" or "determining" or "executing" or "performing" or "collecting" or "creating" or "sending" or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may comprise a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs,

EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

Various general purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description above. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices, etc.

In the foregoing disclosure, implementations of the disclosure have been described with reference to specific example implementations thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of implementations of the disclosure as set forth in the following claims. The disclosure and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A method, comprising:

identifying that a same virtual background was used during video communications associated with multiple meetings based on manual selections, wherein identifying that the same virtual background was used is based on detecting that a user manually selected the same virtual background in each of the multiple meetings;

identifying common information details in respective information details of the multiple meetings;

automatically generating a virtual background selection rule based on identifying that the same virtual background was used during the multiple meetings, wherein the virtual background selection rule associates the common information details of each of the multiple meetings with the same virtual background;

determining that information details of a planned meeting matches the common information details;

in response to determining that the information details of the planned meeting matches the common information details, using the same virtual background of the virtual background selection rule for display during the planned meeting, wherein the same virtual background is used in the planned meeting without user input confirming the same virtual background;

retrieving an image; and

adding the image to the same virtual background.

2. The method of claim 1, further comprising:

retrieving the same virtual background from a virtual background repository for display during the planned meeting.

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3. The method of claim 1, wherein the common information details include at least one of a meeting title, a meeting location, a meeting date and time, or a list of meeting attendees.

4. The method of claim 1, further comprising:
synchronizing the same virtual background across respective devices of at least some users participating in the planned meeting.

5. The method of claim 1, wherein the virtual background selection rule includes key words to match based on Boolean logic operators.

6. The method of claim 1, further comprising:
dynamically changing a virtual background during the planned meeting based on real-time evaluation of meeting content.

7. The method of claim 1, wherein the same virtual background is associated with a user account on a video communication platform.

8. A system, comprising:
a memory; and
a processor, the processor configured to execute instructions stored in the memory to:

identify that a same virtual background was used during video communications associated with multiple meetings based on manual selections, wherein to identify that the same virtual background was used is based on detecting that a user manually selected the same virtual background in each of the multiple meetings;

identify common information details in respective information details of the multiple meetings;

automatically generate a virtual background selection rule based on identifying that the same virtual background was used during the multiple meetings, wherein the virtual background selection rule associates the common information details of each of the multiple meetings with the same virtual background;

determine that information details of a planned meeting matches the common information details;

in response to determining that the information details of the planned meeting matches the common information details, use the same virtual background of the virtual background selection rule for display during the planned meeting, wherein the same virtual background is used in the planned meeting without user input confirming the same virtual background;

retrieve an image; and

add the image to the same virtual background.

9. The system of claim 8, wherein the processor is further configured to execute instructions stored in the memory to: associate the virtual background selection rule with at least one user.

10. The system of claim 8, wherein the processor is further configured to execute instructions stored in the memory to:
dynamically change a virtual background during the planned meeting based on speech-to-text evaluation of meeting content of the planned meeting.

11. The system of claim 8, wherein the virtual background selection rule is based on keywords identified in respective titles of the multiple meetings.

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12. The system of claim 8, wherein the same virtual background is synchronized among multiple user devices connected to the planned meeting.

13. The system of claim 8, wherein the processor is further configured to execute instructions stored in the memory to:

receive an indication to share the virtual background selection rule with a user.

14. A non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising:

identifying that a same virtual background was used during video communications associated with multiple meetings based on manual selections, wherein identifying that the same virtual background was used is based on detecting that a user manually selected the same virtual background in each of the multiple meetings;

identifying common information details in respective information details of the multiple meetings;

automatically generating a virtual background selection rule based on identifying that the same virtual background was used during the multiple meetings, wherein the virtual background selection rule associates the common information details of each of the multiple meetings with the same virtual background;

determining that information details of a planned meeting matches the common information details;

in response to determining that the information details of the planned meeting matches the common information details, using the same virtual background of the virtual background selection rule for display during the planned meeting, wherein the same virtual background is used in the planned meeting without user input confirming the same virtual background;

retrieving an image; and

adding the image to the same virtual background.

15. The non-transitory computer readable medium of claim 14, wherein the same virtual background depicts a logo or company name associated with an email domain.

16. The non-transitory computer readable medium of claim 14, wherein the common information details comprises a common email domain name.

17. The non-transitory computer readable medium of claim 14, wherein the operations further comprise:
storing the virtual background selection rule in association with a user.

18. The non-transitory computer readable medium of claim 14, wherein the operations further comprise:
changing the same virtual background during the planned meeting with another virtual background based on speech-to-text evaluation of meeting content of the planned meeting.

19. The non-transitory computer readable medium of claim 14, wherein the operations further comprise:
transmitting the same virtual background to a device connected to the planned meeting.

20. The non-transitory computer readable medium of claim 14, wherein the operations further comprise:
sharing the virtual background selection rule with a user.

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